Arnab Paul

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SUMMARY

Aerospace R&D engineer with experience in propulsion systems, CFD-based analysis, experimental validation, and thermal-fluid testing for aerospace applications. Delivered engineering work across micro-turbojet performance modelling, droplet-impact testing, pressure/thermal test environments, high-speed imaging diagnostics, and ANSYS Fluent/CFX simulations. Strong fit for open-ended aerospace and defence R&D roles involving first-principles analysis, feasibility studies, concept evaluation, proof-of-concept support, and technical documentation.

CORE ENGINEERING SKILLS

Aerospace & Defence R&D: First-principles analysis, concept evaluation, feasibility studies, system-level thinking, proof-of-concept support, technical documentation

Propulsion Systems: Micro-turbojet performance, liquid propulsion fundamentals, injector concepts, combustor analysis, electric propulsion fundamentals, ion thruster concepts

Thermal-Fluid Engineering: Fluid dynamics, thermodynamics, heat transfer, droplet dynamics, evaporation, heated-wall interaction, multiphase flow

CFD & Simulation: ANSYS Fluent, ANSYS CFX, CFD-Post, VOF modelling, transient simulations, turbulence modelling, reacting-flow analysis, mesh/time-step sensitivity

Experimental Validation: Thermal-pressure test setup, droplet-impact testing, high-speed imaging, heated/non-heated surface testing, atmospheric/elevated-pressure testing

Programming & Analysis: Python, Excel, image-based data extraction, engineering calculations, technical plotting, result interpretation

WORK EXPERIENCE

INDIAN INSTITUTE OF ENGINEERING SCIENCE & TECHNOLOGY — SHIBPUR, INDIA

Project Associate - I — Feb 2024 to May 2026

- Operated a thermal-pressure experimental setup to reproduce propulsion-relevant environments for droplet-impact testing under atmospheric and elevated-pressure conditions.
- Established repeatable test workflows on heated and non-heated surfaces to evaluate spreading, breakup, evaporation response, and surface-interaction behaviour.
- Generated high-speed imaging datasets and converted visual observations into engineering outputs such as spreading ratio, deformation history, breakup tendency, and thermal-response trends.
- Developed VOF-based CFD models in ANSYS Fluent/CFX to simulate droplet impact, interface evolution, evaporation, heat transfer, and wall interaction.
- Improved simulation credibility through mesh-independence and time-step sensitivity checks, followed by comparison with experimental observations.
- Extracted wall heat-flux trends, temperature fields, spreading dynamics, and surface-interaction behaviour for propulsion-relevant thermal-fluid interpretation.
- Converted experimental and CFD outputs into technical reports, plots, presentations, and structured recommendations for next-stage aerospace propulsion R&D.

INDIAN ARMOUR SYSTEMS PRIVATE LIMITED — PALWAL, INDIA

Project Engineer (Aerospace Research & Development) — Feb 2023 to Dec 2023

- Modelled micro-turbojet thrust, mass flow rate, and exhaust gas temperature across altitude-dependent operating conditions up to 10 km.
- Developed a Python-based thermodynamic calculation workflow to automate repeated engine-cycle calculations and accelerate design iteration.
- Conducted aero-thermal performance analysis of thrust output, mass-flow behaviour, exhaust temperature, and altitude response.
- Generated thrust–altitude and thermal-performance plots to identify operating trends and guide design-improvement decisions.

EDUCATION

MASTER'S - PROPULSION ENGINEERING — MOSCOW, RUSSIA

Moscow Aviation Institute — Oct 2020 to Sep 2022

BACHELOR OF TECHNOLOGY - AEROSPACE ENGINEERING — CHENNAI, INDIA

SRM Institute of Science & Technology — Jul 2016 to Jun 2020

PROJECTS

Reduction of CO_2 Emissions Through the Use of Alternative Fuels

Master's Thesis | Moscow Aviation Institute | Feb 2021 – Jun 2022

- Designed and simulated a can-type combustor in ANSYS CFX to compare combustion behaviour, temperature distribution, and emission characteristics across aviation-fuel cases.
- Applied Eddy Dissipation and Finite Rate Chemistry modelling to evaluate reacting-flow behaviour and CO/CO_2 mass-fraction trends.
- Converted injector design, chamber geometry, air-hole placement, temperature contours, and emission outputs into combustor-design recommendations.

Fuel Injector Design and Analysis for a Liquid-Propulsion Combustion Chamber

XLVII International Gagarin Science Conference | Apr 2021

- Developed and evaluated liquid-propulsion injector concepts in ANSYS Fluent to assess pressure conditions, impingement angles, atomization behaviour, and combustion relevance.
- Compared injection configurations to understand spray sensitivity and injector-performance behaviour.
- Secured 2nd Rank at the XLVII International Gagarin Science Conference.

Performance Enhancement of Vertical Axis Wind Turbines Using GA-Based Blade Profile Optimization

B.Tech Major Project | Dec 2019 – Jun 2020

- Combined genetic-algorithm-based airfoil optimisation with RANS-based CFD simulation using MRF/sliding-mesh methods to assess blade-profile performance.

Material Selection for High-Pressure Turbine Blade Applications

B.Tech Minor Project | Jul 2019 – Nov 2019

- Compared Inconel 600, Inconel 754, and Rene 41 under thermal/structural loading and identified Rene 41 as the strongest candidate based on deformation resistance.

PUBLICATIONS

- **Paul, Arnab; Raj, Kanak; Kumar, Pratim; Bhowmik, Joydeep; Lawrence Raj, Prince Raj Numerical Investigation of Droplet Impact Dynamics of Ammonium Dinitramide (ADN)- Based Liquid Propellant on Solid Surfaces Heat Transfer Research (2026)- DOI: 10.1615/HeatTransRes.2026059885**
- **Paul, A., Sarkar, S., Raj, K. *et al.* Droplet Impact Physics of Ammonium Dinitramide (ADN)-Based Monopropellants on Solid Substrates of Varying Roughness. *Fluid Dyn* 61, 4 (2026). https://doi.org/10.1134/S0015462825602013**